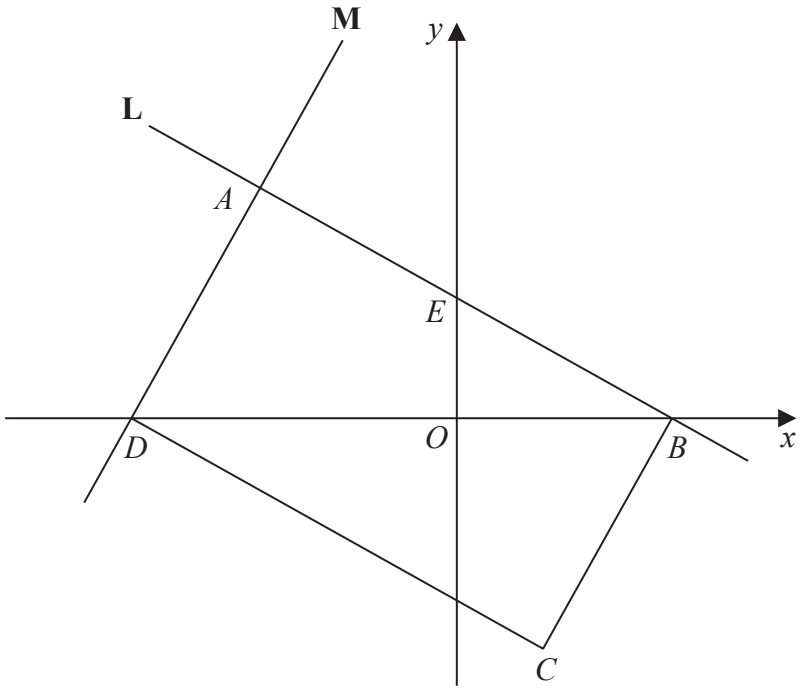
19



ABCD is a rectangle.

A, *E* and *B* are points on the straight line **L** with equation $x + 2y = 12$
A and *D* are points on the straight line **M**.

$AE = EB$

Find an equation for **M**.

L: $y = 6 - \frac{1}{2}x$, so its gradient is $-\frac{1}{2}$.

1 mark
Find the gradient of L

B = (12, 0) and E = (0, 6). E is the midpoint of AB.

A = (2 × 0 − 12, 2 × 6 − 0) = (−12, 12).

Adjacent rectangle sides are perpendicular: gradient of M = 2.

1 mark
Find A and the perpendicular gradient

M: $y = 2x + c$. Substitute A: $12 = 2(-12) + c$.

$c = 12 + 24 = 36$

1 mark
Find the intercept of M

1 mark
Correct equation

$y = 2x + 36$

(Total for Question 19 is 4 marks)

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